

EDGES OF THE UNIFORM RANDOM FOREST OF K_n ARE PAIRWISE NEGATIVELY CORRELATED FOR EVERY n

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ABSTRACT. Let F_n be uniform on all forests of the simple complete graph K_n , with isolated vertices allowed. A conjecture of Kahn and of Winkler, studied by Grimmett and Winkler, asserts that any two distinct edges of any finite graph are negatively correlated under the uniform forest measure. Stark proved this for $G = K_n$ once n is sufficiently large, but did not furnish an explicit threshold. We prove it for every $n \geq 2$, strictly whenever two distinct edges exist. The difficulty is concentrated in the disjoint-edge orbit, whose correlation ratio tends to one. We remove this cancellation before estimating anything: the desired inequality becomes an exact comparison among the first two moments of the component count and the expected sum of squared degrees. When the component count fluctuates, this comparison contains a variance term absent from the fixed-component identities of Tang and Zhang. Component marking, tail elimination, and effective Stirling bounds control the three moments for $n \geq 651$; exact integer recurrences cover the remaining values.

1. INTRODUCTION

Let G be a finite graph and let F be a uniformly random *forest* of G : a uniformly random acyclic subset of the edge set, with isolated vertices permitted, the empty set included, and no connectivity or spanning requirement. Does the presence of one edge in F make a second edge less likely to be present?

For the uniform *spanning tree* the answer is yes, and classically so: the transfer-current theorem makes the edge indicators determinantal, and they are negatively associated. This depends on graphic structure: bases of arbitrary matroids need not even be pairwise negatively correlated [5]. The uniform forest measure is not governed by the spanning-tree determinant, and there the question is open.

Conjecture 1.1 (Kahn’s Conjecture 10.11 [7], Winkler; see Pemantle [9], Grimmett–Winkler [4], Stark [13]). For every finite graph G and every two distinct edges $\varepsilon_1, \varepsilon_2$ of G ,

$$\mathbf{P}(\varepsilon_1 \in F, \varepsilon_2 \in F) \leq \mathbf{P}(\varepsilon_1 \in F) \mathbf{P}(\varepsilon_2 \in F).$$

We call the displayed inequality *pairwise negative correlation* (p-NC). Pemantle [9] records the question as one posed by Winkler. Grimmett and Winkler [4] verified it for all simple graphs on at most 8 vertices and all simple graphs on 9 vertices with at most 18 edges; Semple and Welsh [14] later proved weighted forms for an infinite class of graphs and matroids. For the complete graph, Stark [13] proved p-NC for all sufficiently large n by singularity analysis [3] of the tree and forest generating functions. He records that no explicit value of the eventual threshold is furnished ([13], p. 531); the singularity-analysis remainders carry no numerical constants.

The uniform measure is also the $\beta = 1$ arboreal gas. More generally, activity- β arboreal gas is Bernoulli bond percolation with $p = \beta/(1 + \beta)$ conditioned on acyclicity, and is obtained from the random-cluster model as $q \rightarrow 0$ with $p = \beta q$ [2]. Huang [6] proves negative-correlation results on complete graphs in small- and large- β regimes, but not at

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$\beta = 1$ for every n ; Nguyen and Pylyavskyy [8] give a related combinatorial formula for random-cluster correlations at $q = 1$.

Tang and Zhang [15] prove large- n results for three neighboring uniform measures on K_n : connected spanning subgraphs by disconnection estimates, forests with a fixed number of components from an explicit counting formula and moment inequalities, and connected subgraphs of fixed excess by singularity analysis. Their Remark 1.7 also records that p-NC does not pass between a measure and its truncations, so the fixed-component theorem does not settle the unconditioned forest measure. Their fixed-component moment identities and a later spanning-tree formulation [16] are directly relevant to the reduction used below; the precise comparison is made after Proposition 2.4.

Stark's theorem and the finite census therefore did not meet: without an explicit threshold, no finite computation could close the gap between them. This note supplies a threshold and verifies every case below it.

Throughout, K_n is the **simple** complete graph on the vertex set $[n]$, and F_n is uniform on all forests of K_n .

Theorem 1.2. *For every integer $n \geq 2$ and every pair of distinct edges $e \neq f$ of K_n ,*

$$\mathbf{P}(e \in F_n \text{ and } f \in F_n) \leq \mathbf{P}(e \in F_n) \mathbf{P}(f \in F_n).$$

1.1. Two integer inequalities. Because $\text{Aut}(K_n) = S_n$ is transitive on edges and on each of the two orbits of pairs of distinct edges, Theorem 1.2 is equivalent to a pair of inequalities between integers. Write

$$A(n) = \#\{\text{forests of } K_n\}, \quad N_e(n), N_{\text{adj}}(n), N_{\text{dis}}(n)$$

for the number of forests containing, respectively, one fixed edge, a fixed pair of adjacent edges, and a fixed pair of disjoint edges. Then Theorem 1.2 says exactly

$$(\text{ADJ}_n) \quad N_{\text{adj}}(n) A(n) \leq N_e(n)^2 \quad (n \geq 3),$$

$$(\text{DIS}_n) \quad N_{\text{dis}}(n) A(n) \leq N_e(n)^2 \quad (n \geq 4).$$

The cases $n = 2$ (only one edge) and, for the disjoint orbit, $n = 3$ (no disjoint pair) are vacuous.

In fact both inequalities are strict in every nonvacuous finite case; the disjoint ratio merely tends to 1.

The first instance with both orbits already shows the difference between them. For $n = 4$ one has $(A, N_e, N_{\text{adj}}, N_{\text{dis}}) = (38, 14, 4, 5)$, so the joint-to-product ratios are $38/49$ for an adjacent pair and $95/98$ for a disjoint pair.

1.2. Why the disjoint inequality is delicate. The two orbits are not of equal difficulty, and the reason is visible already in the spanning-tree stratum of the forest measure.

Observation 1.3 (Stark, Theorems 1.2 and 1.3 [13]). Let T be a uniform spanning tree of K_n , $n \geq 3$. Then $\mathbf{P}(\varepsilon \in T) = 2/n$; adjacent pairs satisfy $\mathbf{P}(\varepsilon_1, \varepsilon_2 \in T) = 3n^{-2}$; and nonadjacent pairs satisfy $\mathbf{P}(\varepsilon_1, \varepsilon_2 \in T) = 4n^{-2} = \mathbf{P}(\varepsilon_1 \in T)\mathbf{P}(\varepsilon_2 \in T)$, i.e. they are exactly independent. Theorem 1.3 of [13] is stronger still: for $n \geq 2s$, any s mutually nonadjacent edges of K_n are jointly independent in T .

Both statements also follow at once from Lemma 2.2 below, which gives $2^s n^{n-s-2}$ spanning trees containing s prescribed disjoint edges.

For an adjacent pair the spanning tree already has a definite $3/4$ of slack, and (ADJ_n) inherits it. For a disjoint pair the spanning tree sits exactly at equality, so no slack whatever is available from that stratum, and what survives in (DIS_n) is correspondingly thin. Stark quantified it.

Observation 1.4 (Stark [13], Theorem 1.4). For the uniform forest of K_n , $\mathbf{P}(\varepsilon_1 \in F)\mathbf{P}(\varepsilon_2 \in F) = 4n^{-2} - 4n^{-3} - 23n^{-4} + O(n^{-5})$ and, for nonadjacent pairs, $\mathbf{P}(\varepsilon_1, \varepsilon_2 \in F)$

$F) = 4n^{-2} - 4n^{-3} - 27n^{-4} + O(n^{-5})$. Dividing, the disjoint ratio is $1 - n^{-2} + O(n^{-3})$. For an adjacent pair the same theorem gives $\mathbf{P}(\varepsilon_1, \varepsilon_2 \in F) = 3n^{-2} + O(n^{-3})$, so its ratio tends to $3/4$.

So (DIS_n) is asymptotically tight, with relative slack of order n^{-2} only. Any argument that loses $\Theta(n^{-2})$ at any point cannot close the disjoint orbit. The initial reduction must therefore be exact, and the subsequent estimates must preserve the leading terms.

1.3. Outline of the proof. The adjacent orbit is settled quickly (Section 7): comparing the component-peeling recurrences for N_{adj} and N_e term by term gives $2(n-2)N_{\text{adj}} \leq 3N_e$, and (ADJ_n) then follows from any bound $\mathbf{E}[c_n] \leq 2$ on the mean number of components. The disjoint orbit is where the work lies, and the argument has three stages.

Let c be the number of components of F_n , let $m = n - c$ be its number of edges, and let $D = \sum_v \deg_{F_n}(v)^2$. Double counting expresses the three edge counts through moments of m and D . Proposition 3.1 then makes (DIS_n) equivalent to $\Delta(n) \geq 0$, an inequality involving only $\mathbf{E}[D]$, $\mathbf{E}[c]$, and $\mathbf{E}[c^2]$. Tang and Zhang obtained the corresponding fixed- m identities [15, 16]; here m fluctuates, and the new term $\text{Var}(m)$ must be controlled. The equivalence removes the order- n^{-2} cancellation before any estimate is made.

Component-marking identities reduce the needed bounds on $\mathbf{E}[c]$ and $\mathbf{E}[c^2]$ to effective control of $a_n := A(n)/n^{n-2}$. Robbins' form of Stirling's inequality supplies that control. A tail-elimination step then absorbs all but forty explicit terms of the second-moment identity into the already bounded mean.

Finally, expanding $\sum_F D(F)$ over the component of a fixed vertex gives $\mathbf{E}[D_n] \geq 5n - \frac{29}{2}$. The only inputs to the estimates from the literature are Prüfer's bijection and Robbins' inequality; Appendix A records the exact rational verification of every constant.

1.4. Scope, and the role of computation. Theorem 1.2 concerns the single family $G = K_n$ of simple graphs and the pairwise form of the conjecture. Conjecture 1.1 for general finite graphs remains open, as do its multigraph form — the one tied to the pairwise $q < 1$ random-cluster question — full negative association of the uniform forest measure, even for K_n , and the connected-subgraph analogue proposed in [4]. The measure here is uniform on *all* forests: it is not the uniform spanning forest of the probabilistic literature, not the uniform spanning tree, and not a forest conditioned to have a prescribed number of components. Nothing below improves the constant $c = 2$ in the universal bound $\mathbf{P}(\varepsilon_1, \varepsilon_2 \in F) \leq c\mathbf{P}(\varepsilon_1)\mathbf{P}(\varepsilon_2)$ valid for all graphs [1].

Finite ranges are checked in exact integers. Each analytic estimate ends in a rational expression with one-sided enclosures for e , π , $\sqrt{2\pi}$, and $\zeta(3/2)$, so no floating-point value decides an inequality. The program `tests/check.py` and its limits are described in Appendix A; universal inequalities such as $\log(1-x) \leq -x - x^2/2$ are proved in the text, not delegated to computation.

2. NOTATION, EXACT COUNTS AND IDENTITIES

2.1. Notation.

$A(n)$	number of forests of K_n ; $A(0) = A(1) = 1$ (OEIS A001858, [17])
a_n	$A(n)/n^{n-2}$, with $1^{-1} = 1$ and $2^0 = 1$, so $a_1 = 1$, $a_2 = 2$
$t(k)$	k^{k-2} , the number of trees on a k -set; $t(1) = t(2) = 1$
$c = c(F)$	number of components of the forest F , isolated vertices counted
$m = F $	number of edges of F ; always $m = n - c$
$D = D(F)$	$\sum_{v \in [n]} \deg_F(v)^2$
$\mathbf{E}$	expectation with respect to the uniform measure on the $A(n)$ forests
μ	$\mathbf{E}[c_n]$
$N_e, N_{\text{adj}}, N_{\text{dis}}$	as in the introduction

Lemma 2.1 (Prüfer [10]). *For $k \geq 2$ there is a bijection between trees on $[k]$ and sequences in $[k]^{k-2}$ under which $\deg_T(v) = 1 + (\text{number of occurrences of } v)$ for every v . Consequently the number of trees on $[k]$ is $t(k) = k^{k-2}$, and the number with prescribed degrees $d_1, \dots, d_k$ (all ≥ 1 , summing to $2k - 2$) is $\binom{k-2}{d_1-1, \dots, d_k-1}$.*

2.2. Trees through a prescribed forest. Every count in this note is assembled by peeling off one component of a forest, so we first need to know how many trees contain a given forest.

Lemma 2.2. *Let $S_1, \dots, S_r$ be a partition of $[k]$ with $|S_i| = s_i$, and let Φ be a forest on $[k]$ whose components have vertex sets exactly $S_1, \dots, S_r$. Then the number of trees T on $[k]$ with $\Phi \subseteq T$ is*

$$k^{r-2} \prod_{i=1}^r s_i,$$

read as 1 when $r = 1$.

Proof. A tree $T \supseteq \Phi$ is obtained by adding $r - 1$ edges joining distinct S_i 's so that the contracted multigraph on $\{1, \dots, r\}$ is a tree. For $r = 1$ there is nothing to add. For $r \geq 2$, contract each S_i to a node i : the added edges form a tree τ on $[r]$, and if node i has degree d_i in τ then the d_i endpoints at i may be chosen independently among the s_i vertices of S_i , in $s_i^{d_i}$ ways. Distinct choices give distinct trees T , and every $T \supseteq \Phi$ arises once. By Lemma 2.1 the number of τ with degree sequence (d_i) is $(r - 2)! / \prod_i (d_i - 1)!$, so, writing $e_i = d_i - 1 \geq 0$ with $\sum_i e_i = r - 2$,

$$\begin{aligned} \#\{T \supseteq \Phi\} &= \sum_{(d_i)} \frac{(r - 2)!}{\prod_i (d_i - 1)!} \prod_i s_i^{d_i} \\ &= \left(\prod_i s_i \right) \sum_{\substack{e_i \geq 0 \\ \sum e_i = r - 2}} \frac{(r - 2)!}{\prod_i e_i!} \prod_i s_i^{e_i} \\ &= \left(\prod_i s_i \right) \left(\sum_i s_i \right)^{r-2}, \end{aligned}$$

by the multinomial theorem, and $\sum_i s_i = k$. $\square$

Applying Lemma 2.2 to Φ consisting of one edge, of two adjacent edges, and of two disjoint edges (plus isolated vertices) gives the three counts we need:

$$(1) \quad t_2(k) = 2k^{k-3} \ (k \geq 2), \quad t_3(k) = 3k^{k-4} \ (k \geq 3), \quad t_4(k) = 4k^{k-4} \ (k \geq 4),$$

being the number of trees on a k -set containing a fixed edge, a fixed pair of adjacent edges, and a fixed pair of disjoint edges respectively. The degenerate values are $t_2(2) = 1$, $t_3(3) = 1$, $t_4(4) = 4$, consistent with the rational reading of the powers.

2.3. Peeling recurrences.

Proposition 2.3. For $n \geq 1$,

$$(2) \quad A(n) = \sum_{k=1}^n \binom{n-1}{k-1} t(k) A(n-k), \quad A(0) = 1,$$

$$(3) \quad N_e(n) = \sum_{k=2}^n \binom{n-2}{k-2} t_2(k) A(n-k),$$

$$(4) \quad N_{\text{adj}}(n) = \sum_{k=3}^n \binom{n-3}{k-3} t_3(k) A(n-k),$$

$$(5) \quad N_{\text{dis}}(n) = \sum_{k=4}^n \binom{n-4}{k-4} t_4(k) A(n-k) + \sum_{k=2}^{n-2} \binom{n-4}{k-2} t_2(k) N_e(n-k).$$

Proof. For (2), classify forests by the vertex set $S \ni 1$ of the component containing vertex 1: there are $\binom{n-1}{k-1}$ choices of S with $|S| = k$, $t(k)$ trees on S , and $A(n-k)$ forests on $[n] \setminus S$. For (3) and (4), do the same with S the vertex set of the component containing the required edge, resp. the required adjacent pair, using (1). For (5), split according to whether the two required disjoint edges lie in one component or in two. In the first case S is that component and (1) applies. In the second case, let S be the component containing the *first* of the two edges: it contains that edge and avoids the two endpoints of the second, giving $\binom{n-4}{k-2}$ choices and $t_2(k)$ trees, and the remaining $n-k$ vertices carry a forest containing the second edge, of which there are $N_e(n-k)$. The two components are distinguished by which required edge they contain, so no factor $1/2$ occurs. $\square$

2.4. Moment identities. The next proposition is what converts the three pair counts into moments; it is the entry point for the reduction of Section 3.

Proposition 2.4. For every $n \geq 4$,

$$(6) \quad n(n-1) \frac{N_e(n)}{A(n)} = 2 \mathbf{E}[m],$$

$$(7) \quad n(n-1)(n-2) \frac{N_{\text{adj}}(n)}{A(n)} = \mathbf{E}[D] - 2 \mathbf{E}[m],$$

$$(8) \quad n(n-1)(n-2)(n-3) \frac{N_{\text{dis}}(n)}{A(n)} = 4(\mathbf{E}[m^2] + \mathbf{E}[m] - \mathbf{E}[D]).$$

Proof. Fix a forest F with m edges. Since no two edges of a forest share two vertices, $\sum_v \deg_F(v)^2$ counts ordered pairs (e, f) of edges of F with a common vertex, with $e = f$ allowed; the diagonal contributes $\sum_v \deg_F(v) = 2m$. Hence F has $D - 2m$ ordered pairs of distinct adjacent edges and $m(m-1) - (D - 2m) = m^2 + m - D$ ordered pairs of disjoint edges.

Now sum over all forests. K_n has $\binom{n}{2}$ edges, $n(n-1)(n-2)$ ordered pairs of distinct adjacent edges (choose the shared vertex, then the two other endpoints in order), and $n(n-1)(n-2)(n-3)/4$ ordered pairs of disjoint edges. By edge- and orbit-transitivity of $\text{Aut}(K_n)$,

$$\begin{aligned} \sum_F m &= \binom{n}{2} N_e(n), & \sum_F (D - 2m) &= n(n-1)(n-2) N_{\text{adj}}(n), \\ \sum_F (m^2 + m - D) &= \frac{1}{4} n(n-1)(n-2)(n-3) N_{\text{dis}}(n). \end{aligned}$$

Dividing by $A(n)$ gives (6)–(8). $\square$

When the measure is supported on forests with a fixed number of components, m is deterministic and (6)–(8) specialize to the identities of Tang and Zhang [15, Lemma 3.2]; their Corollary 3.3 gives the corresponding p-NC criteria. Their later Lemma 2.1 [16] places

the spanning-tree version in a broader graph setting. Proposition 2.4 is used here for the unconditioned forest, where m varies; the variance term created by that variation is explicit in the next section.

Since $m = n - c$, (6) reads

$$(9) \quad \frac{N_e(n)}{A(n)} = \frac{n - \mu}{\binom{n}{2}}, \quad \mu = \mathbf{E}[c_n].$$

Finally we record the peeling identity for the degree moment, needed in Section 6.

Proposition 2.5. *Let $g(k) := \sum_T \deg_T(1)^2$, the sum over all trees T on $[k]$. Then $g(1) = 0$, $g(2) = 1$, and for $k \geq 2$*

$$(10) \quad g(k) = k^{k-2} + 3(k-2)k^{k-3} + (k-2)(k-3)k^{k-4},$$

(so $g(3) = 6$), and for $n \geq 1$

$$(11) \quad \sum_F D(F) = n \sum_{k=1}^n \binom{n-1}{k-1} g(k) A(n-k).$$

Proof. For $k \geq 2$ apply Lemma 2.1: under the uniform measure on the k^{k-2} trees, $\deg_T(1) = 1 + X$ where X is the number of occurrences of the label 1 in a uniform word of $[k]^{k-2}$, so $X \sim \text{Bin}(k-2, 1/k)$. Then $\mathbf{E}[X] = (k-2)/k$ and $\mathbf{E}[X^2] = (k-2)(k-1)/k^2 + (k-2)^2/k^2 = (k-2)(2k-3)/k^2$, whence

$$\begin{aligned} g(k) &= k^{k-2} \mathbf{E}[(1+X)^2] = k^{k-2} + 2(k-2)k^{k-3} + (k-2)(2k-3)k^{k-4} \\ &= k^{k-2} + 3(k-2)k^{k-3} + (k-2)(k-3)k^{k-4}, \end{aligned}$$

the last equality because both middle-plus-last groups equal $(k-2)(4k-3)k^{k-4}$. For (11), vertex-transitivity gives $\sum_F D(F) = n \sum_F \deg_F(1)^2$, and peeling the component of vertex 1 as in (2) evaluates $\sum_F \deg_F(1)^2$ as $\sum_k \binom{n-1}{k-1} g(k) A(n-k)$. $\square$

3. THE EXACT REDUCTION OF THE DISJOINT ORBIT

The moment identities make (DIS_n) equivalent to the following inequality; every step in the reduction is reversible.

Proposition 3.1. *For every $n \geq 4$, put*

$$\Delta(n) := \mathbf{E}[D] - \mathbf{E}[m] - \text{Var}(m) - \frac{(4n-6) \mathbf{E}[m]^2}{n(n-1)}.$$

Then

$$(\text{DIS}_n) \iff \Delta(n) \geq 0,$$

with equality in the disjoint inequality exactly when $\Delta(n) = 0$. Moreover, writing $\mu = \mathbf{E}[c_n]$ (so $\mathbf{E}[m] = n - \mu$ and $\text{Var}(m) = \text{Var}(c)$),

$$(12) \quad \Delta(n) = (\mathbf{E}[D] - 5n) + 2 + 9\mu + \mu^2 - \mathbf{E}[c^2] - R(n, \mu), \quad R(n, \mu) = \frac{(4\mu^2 + 4\mu - 2)n - 6\mu^2}{n(n-1)}.$$

If moreover $1 \leq \mu \leq 2$ then $0 < R(n, \mu) \leq 22/(n-1)$.

Proof. Both $A(n)$ and $N_e(n)$ are positive for $n \geq 4$, so (DIS_n) is equivalent to $N_{\text{dis}}/A \leq (N_e/A)^2$. By (6) and (8) this reads

$$\frac{4(\mathbf{E}[m^2] + \mathbf{E}[m] - \mathbf{E}[D])}{n(n-1)(n-2)(n-3)} \leq \frac{4\mathbf{E}[m]^2}{n^2(n-1)^2},$$

i.e., after multiplying by the positive quantity $n(n-1)(n-2)(n-3)/4$ and rearranging,

$$\mathbf{E}[D] \geq \mathbf{E}[m^2] + \mathbf{E}[m] - \frac{(n-2)(n-3)}{n(n-1)} \mathbf{E}[m]^2.$$

Substituting $\mathbf{E}[m^2] = \text{Var}(m) + \mathbf{E}[m]^2$ and $1 - \frac{(n-2)(n-3)}{n(n-1)} = \frac{4n-6}{n(n-1)}$ gives $\Delta(n) \geq 0$, and every step is reversible.

For (12), divide $(4n-6)(n-\mu)^2$ by $n(n-1)$:

$$(4n-6)(n-\mu)^2 = (4n-8\mu-2)n(n-1) + [(4\mu^2+4\mu-2)n-6\mu^2],$$

which is checked by expanding both sides. Hence

$$\Delta = \mathbf{E}[D] - (n-\mu) - (\mathbf{E}[c^2] - \mu^2) - (4n-8\mu-2+R) = (\mathbf{E}[D] - 5n) + 2 + 9\mu + \mu^2 - \mathbf{E}[c^2] - R.$$

Finally, if $1 \leq \mu \leq 2$ then $4\mu^2 + 4\mu - 2 \geq 6 > 0$, so $0 < R \leq \frac{(4\mu^2+4\mu-2)n}{n(n-1)} = \frac{4\mu^2+4\mu-2}{n-1} \leq \frac{22}{n-1}$. $\square$

Thus it is enough to control $\mathbf{E}[D] - 5n$, $\mathbf{E}[c]$, and $\mathbf{E}[c^2]$ to constant accuracy. There is little room: the final margin at $n = 651$ is 0.0870 with the proved second-moment bound, and falls below 0.01 if that input is rounded up to 2.99. The numerical losses still matter.

4. EFFECTIVE BOUNDS FOR THE FOREST COUNT

The moment estimates all pass through the normalised forest count $a_n = A(n)/n^{n-2}$. Stark's expansion [13, Lemma 4.1] gives $a_n = \sqrt{e}(1 + 5/(2n) + 11/(8n^2)) + O(n^{-3})$, refining the classical limit $a_n \rightarrow \sqrt{e}$ [11]; its unspecified remainder is not an effective two-sided bound. This section proves $1.647767 \leq a_n \leq 17/10$ for all $n \geq 81$, together with the crude $a_n \leq 5/2$ valid for all n .

We use throughout the following two elementary facts about the logarithm: for $0 \leq x < 1$,

$$(13) \quad -x - \frac{x^2}{2(1-x)} \leq \log(1-x) \leq -x - \frac{x^2}{2} \leq -x.$$

(The upper bound is the truncation of the power series $\log(1-x) = -\sum_{i \geq 1} x^i/i$ after two terms, all discarded terms being negative; the lower bound follows from $\sum_{i \geq 2} x^i/i \leq \frac{1}{2} \sum_{i \geq 2} x^i = \frac{x^2}{2(1-x)}$.) Together with Robbins' sharpening of Stirling's formula these give everything we need about binomial coefficients.

Lemma 4.1 (Robbins [12]). *For every integer $s \geq 1$,*

$$\sqrt{2\pi s} \left(\frac{s}{e}\right)^s e^{\frac{1}{12s+1}} \leq s! \leq \sqrt{2\pi s} \left(\frac{s}{e}\right)^s e^{\frac{1}{12s}}.$$

Lemma 4.2 (binomial bound). *For all integers $1 \leq k \leq n-1$,*

$$\binom{n}{k} \leq \frac{n^{n+1/2}}{\sqrt{2\pi} k^{k+1/2} (n-k)^{n-k+1/2}}.$$

Proof. Apply the upper bound of Lemma 4.1 to $n!$ and the lower bound to $k!$ and $(n-k)!$. The factors e^{-n} cancel, $\sqrt{2\pi n}/(\sqrt{2\pi k}\sqrt{2\pi(n-k)}) = \frac{1}{\sqrt{2\pi}}\sqrt{n/(k(n-k))}$, and the residual exponential is $\exp(\frac{1}{12n} - \frac{1}{12k+1} - \frac{1}{12(n-k)+1})$. Since $k \leq n-1$ we have $\frac{1}{12k+1} \geq \frac{1}{12n-11} > \frac{1}{12n}$, so that exponent is negative and the factor is < 1 . $\square$

Re-indexing (2) by the size $m = n - k$ of the complement of the peeled component and dividing by n^{n-2} gives the recursion that drives this section:

$$(14) \quad a_n = 1 + \sum_{m=1}^{n-1} q_m(n) a_m \quad (n \geq 1),$$

where, for $1 \leq m \leq n-1$,

$$(15) \quad q_m(n) := \frac{\binom{n-1}{m} m^{m-2} (n-m)^{n-m-2}}{n^{n-2}}, \quad w_m := \frac{m^{m-2} e^{-m}}{m!}, \quad v_m := \frac{A(m) e^{-m}}{m!},$$

and the leading 1 in (14) is the term $m = 0$. The kernel $q_m(n)$ must be controlled in two different regimes: crudely but uniformly for large m , and sharply for small m , where the mass sits.

Lemma 4.3 (Stirling bound on the kernel). *For all integers $1 \leq m \leq n-1$, with $k = n-m$,*

$$q_m(n) \leq \frac{n^{3/2}}{\sqrt{2\pi} m^{5/2} k^{3/2}}.$$

Proof. Since $(n-1)! = n!/n$ and $(k-1)! = k!/k$, we have $\binom{n-1}{m} = \frac{(n-1)!}{m!(k-1)!} = \frac{k}{n} \binom{n}{m}$. By Lemma 4.2,

$$q_m(n) = \frac{k}{n} \binom{n}{m} \frac{m^{m-2} k^{k-2}}{n^{n-2}} \leq \frac{k}{n} \cdot \frac{n^{n+1/2}}{\sqrt{2\pi} m^{m+1/2} k^{k+1/2}} \cdot \frac{m^{m-2} k^{k-2}}{n^{n-2}} = \frac{n^{3/2}}{\sqrt{2\pi} m^{5/2} k^{3/2}}.$$

□

For small m the right comparison is with w_m , the $n \rightarrow \infty$ limit of the kernel. Write

$$(16) \quad R_m(n) := \frac{q_m(n)}{w_m} = e^m \prod_{i=1}^m \left(1 - \frac{i}{n}\right) \left(1 - \frac{m}{n}\right)^{n-m-2}, \quad \rho_m(n) := \frac{R_m(n)}{1 - m/n},$$

the displayed form of q_m/w_m following from $\binom{n-1}{m} m! = (n-1)(n-2) \cdots (n-m) = n^m \prod_{i=1}^m (1 - i/n)$.

Lemma 4.4 (upper estimate for the kernel ratio). *For $1 \leq m \leq n-2$,*

$$R_m(n) \leq \exp\left(\frac{3m}{2n} + \frac{m^2(m+2)}{2n^2}\right), \quad \rho_m(n) \leq \exp\left(\frac{5m}{2n} + \frac{m^3}{2n^2} + \frac{m^2}{n^2}\right).$$

Proof. For the second bound, $\log \rho_m = m + \sum_{i=1}^{m-1} \log(1 - i/n) + (n-m-2) \log(1 - m/n)$. By (13), $\sum_{i=1}^{m-1} \log(1 - i/n) \leq -\frac{m(m-1)}{2n}$, and, using $n-m-2 \geq 0$,

$$(n-m-2) \log\left(1 - \frac{m}{n}\right) \leq (n-m-2) \left(-\frac{m}{n} - \frac{m^2}{2n^2}\right) = -m + \frac{m^2}{n} + \frac{2m}{n} - \frac{m^2}{2n} + \frac{m^3}{2n^2} + \frac{m^2}{n^2}.$$

Adding, the coefficient of m^2/n is $-\frac{1}{2} + 1 - \frac{1}{2} = 0$ and the coefficient of m/n is $\frac{1}{2} + 2 = \frac{5}{2}$, giving the stated bound. The first bound is identical except that the product runs to $i = m$, contributing $-\frac{m(m+1)}{2n}$ instead of $-\frac{m(m-1)}{2n}$; the coefficient of m/n becomes $-\frac{1}{2} + 2 = \frac{3}{2}$, and $\frac{m^3}{2n^2} + \frac{m^2}{n^2} = \frac{m^2(m+2)}{2n^2}$. □

Lemma 4.5 (lower estimate for the kernel ratio). *For $1 \leq m \leq n-2$,*

$$R_m(n) \geq \exp\left(\frac{3m}{2n} - \frac{(m+1)^3}{6n(n-m)}\right),$$

so that $R_m(n) \geq 1$, and hence $\rho_m(n) \geq R_m(n) \geq 1$, whenever

$$(17) \quad 9m(n-m) \geq (m+1)^3.$$

Proof. By the left inequality of (13),

$$\sum_{i=1}^m \log\left(1 - \frac{i}{n}\right) \geq -\frac{m(m+1)}{2n} - \sum_{i=1}^m \frac{i^2}{2n(n-i)} \geq -\frac{m(m+1)}{2n} - \frac{m(m+1)(2m+1)}{12n(n-m)},$$

and, since $0 \leq \frac{n-m-2}{n-m} \leq 1$,

$$(n-m-2) \log\left(1 - \frac{m}{n}\right) \geq (n-m-2) \left(-\frac{m}{n} - \frac{m^2}{2n(n-m)}\right) \geq -m + \frac{m^2}{n} + \frac{2m}{n} - \frac{m^2}{2n}.$$

Adding m and collecting, the m^2/n terms cancel as before and

$$\log R_m(n) \geq \frac{3m}{2n} - \frac{m(m+1)(2m+1)}{12n(n-m)}.$$

Finally $\frac{m(m+1)(2m+1)}{12} \leq \frac{(m+1)^3}{6}$, because after dividing by $(m+1)/6$ it reads $m(2m+1)/2 \leq (m+1)^2$, i.e. $2m^2 + m \leq 2m^2 + 4m + 2$. The exponent is nonnegative exactly when $\frac{3m}{2n} \geq \frac{(m+1)^3}{6n(n-m)}$, i.e. when (17) holds. $\square$

The lower bound on a_n is now immediate: on the range where $R_m(n) \geq 1$, the recursion (14) dominates its own limiting form term by term.

Theorem 4.6 (lower bound, all n). $a_n \geq 1.647767$ for every $n \geq 2$.

Proof. For fixed n , the difference $9m(n-m) - (m+1)^3$ is concave in m , so its minimum on $1 \leq m \leq 60$ is at an endpoint. The $m = 1$ endpoint is positive here, while the $m = 60$ endpoint reads $9 \cdot 60 \cdot (n-60) \geq 61^3 = 226981$, i.e. $n \geq 480.335185\dots$; so it holds for every $n \geq 481$ and fails at $n = 480$ ($9 \cdot 60 \cdot 420 = 226800 < 226981$). Note that $w_m a_m = \frac{m^{m-2} e^{-m}}{m!} \cdot \frac{A(m)}{m^{m-2}} = v_m$, so $q_m(n) a_m = w_m R_m(n) a_m = v_m R_m(n) \geq v_m$ by Lemma 4.5. Hence, for $n \geq 481$, dropping every term of (14) with $m > 60$ (all terms are positive),

$$a_n \geq 1 + \sum_{m=1}^{60} q_m(n) a_m \geq 1 + \sum_{m=1}^{60} v_m = 1 + \sum_{m=1}^{60} \frac{A(m)}{e^m m!} = 1.647767048\dots > 1.647767.$$

For $2 \leq n \leq 500$ the inequality is verified in exact integers. The two ranges overlap. $\square$

The upper bound needs more care, because the recursion has to be closed on itself; we induct, splitting the sum at $m = 60$ and at $m = n/2$.

Theorem 4.7 (upper bound). $a_n \leq 17/10$ for every $n \geq 81$, and the threshold 81 is sharp: $a_{80} = 1.700555369\dots > 17/10$.

Proof. *Exact range.* $a_{80} > 17/10$ and $a_n \leq 17/10$ for $81 \leq n \leq 500$ are exact integer computations.

Analytic range. We show by strong induction that $a_n \leq 17/10$ for every $n \geq 500$; the base $n = 500$ is in the exact range, and in the inductive step we may use $a_r \leq 17/10$ for $81 \leq r < n$ and the exact table for $r \leq 500$. Split (14) at $m = 60$ and $m = n/2$.

Head, $1 \leq m \leq 60$. By Lemma 4.4, $q_m(n) a_m = v_m R_m(n) \leq v_m \exp(\frac{3m}{2n} + \frac{m^2(m+2)}{2n^2})$, and each factor decreases in n , so with $n \geq 500$,

$$\sum_{m=1}^{60} q_m(n) a_m \leq \sum_{m=1}^{60} v_m \exp\left(\frac{3m}{1000} + \frac{m^2(m+2)}{500000}\right) \leq 0.652836237.$$

Middle, $60 < m \leq n/2$. Here $n-m \geq n/2$, so Lemma 4.3 gives $q_m(n) \leq 2^{3/2}/(\sqrt{2\pi} m^{5/2})$, uniformly in n . On $61 \leq m \leq 80$ use the exact bound $a_m \leq 5/2$ (the maximum of a_m over $1 \leq m \leq 80$ is $a_4 = 19/8$); on $m \geq 81$ use the induction hypothesis $a_m \leq 17/10$ (legitimate since $m < n$). With $\sum_{m=a+1}^b m^{-5/2} \leq \int_a^b x^{-5/2} dx = \frac{2}{3}(a^{-3/2} - b^{-3/2})$,

$$\sum_{60 < m \leq n/2} q_m(n) a_m \leq \frac{2^{3/2}}{\sqrt{2\pi}} \left(\frac{5}{2} \sum_{61 \leq m \leq 80} m^{-5/2} + \frac{17}{10} \sum_{m \geq 81} m^{-5/2} \right) \leq 0.00320544,$$

a bound independent of n .

Far, $n/2 < m \leq n-1$. Here $m \geq n/2 \geq 250 \geq 81$, so the induction hypothesis gives $a_m \leq 17/10$ throughout, and Lemma 4.3 with $m^{-5/2} \leq 2^{5/2} n^{-5/2}$ gives, writing $k = n-m$,

$$\begin{aligned} \sum_{n/2 < m \leq n-1} q_m(n) a_m &\leq \frac{17}{10} \cdot \frac{2^{5/2}}{\sqrt{2\pi} n} \sum_{k \geq 1} k^{-3/2} \\ &\leq \frac{17}{10} \cdot \frac{2^{5/2} \zeta(3/2)}{\sqrt{2\pi} n} \leq 0.020044700 \quad (n \geq 500). \end{aligned}$$

Summing, $a_n \leq 1 + 0.652836237 + 0.00320544 + 0.020044700 = 1.676086377 \leq 17/10$ for $n = 500$; the head majorant decreases in n , the middle majorant is independent of n , and the far majorant is proportional to $1/n$, so the same bound holds for every $n \geq 500$. $\square$

Corollary 4.8. $a_n \leq 5/2$ for every $n \geq 1$, with equality nowhere; the maximum is $a_4 = 19/8$.

Proof. Exact integers for $1 \leq n \leq 80$, and Theorem 4.7 for $n \geq 81$. $\square$

In the proof of Theorem 4.7, the bound $a_m \leq 5/2$ on $61 \leq m \leq 80$ comes from the exact table, independently of this corollary.

5. EFFECTIVE MOMENTS OF THE COMPONENT COUNT

We turn to $\mathbf{E}[c_n]$ and $\mathbf{E}[c_n^2]$. The tool is a pair of identities obtained by marking one component, or an ordered pair of components, of a random forest.

Lemma 5.1 (component marking). *For $n \geq 1$ set*

$$\tau_j(n) := \frac{\binom{n}{j}(n-j)^{n-j-2}A(j)}{A(n)} \quad (0 \leq j \leq n-1),$$

$$\sigma_k(n) := \frac{\binom{n-1}{k}k^{k-2}A(n-k)}{A(n)} \quad (1 \leq k \leq n-1).$$

Then

$$(18) \quad \sum_{j=0}^{n-1} \tau_j(n) = \mathbf{E}[c_n], \quad \tau_0(n) = \frac{1}{a_n},$$

$$(19) \quad \mathbf{E}[c_n(c_n - 1)] = \sum_{j=1}^{n-1} \tau_j(n) \mathbf{E}[c_j] \quad (n \geq 2),$$

$$(20) \quad \mathbf{E}[c_n] = 1 + \sum_{k=1}^{n-1} \sigma_k(n),$$

and moreover

$$(21) \quad \sigma_k(n) = q_k(n) \frac{a_{n-k}}{a_n} = w_k \left(1 - \frac{k}{n}\right) \rho_k(n) \frac{a_{n-k}}{a_n}, \quad \tau_j(n) = \rho_j(n) \frac{v_j}{a_n}.$$

Proof. (18): count pairs (F, C) with C a component of F , classifying by the vertex set of C ; if $|C| = n - j$ there are $\binom{n}{j}$ choices of the complement, $(n - j)^{n-j-2}$ trees on C and $A(j)$ forests on the complement. Dividing by $A(n)$ gives $\sum_j \tau_j = \mathbf{E}[c]$. The term $j = 0$ is $n^{n-2}/A(n) = 1/a_n$.

(19): the same count with an *ordered* pair of distinct components (C_1, C_2) . Choose C_1 as above with complement of size j ; C_2 is then any component of the forest induced on the complement, and summing over those forests gives $A(j)\mathbf{E}[c_j]$. The term $j = 0$ contributes 0.

(20): set

$$\text{term}_k := \binom{n}{k} k^{k-2} A(n-k) / A(n) = \tau_{n-k}.$$

Then (18) reads $\mathbf{E}[c_n] = \sum_{k=1}^n \text{term}_k$. Counting pairs consisting of a component and one of its vertices gives

$$\sum_{k=1}^n k \text{term}_k = n, \quad \sum_k \frac{k}{n} \text{term}_k = 1.$$

Subtracting leaves $\mathbf{E}[c_n] - 1 = \sum_{k=1}^{n-1} (1 - k/n) \text{term}_k$; the $k = n$ term vanishes. Finally $(1 - k/n) \binom{n}{k} = \binom{n-1}{k}$ identifies the summand with $\sigma_k(n)$.

(21): the first equality is (15) together with $A(r) = a_r r^{r-2}$, and the second is (16). For the third, use $\binom{n}{j} j! = n(n-1) \cdots (n-j+1) = n^j \prod_{i=1}^{j-1} (1 - i/n)$ to get

$$\tau_j(n) = \prod_{i=1}^{j-1} \left(1 - \frac{i}{n}\right) \left(1 - \frac{j}{n}\right)^{n-j-2} \cdot \frac{A(j)}{j!} \cdot \frac{n^{n-2}}{A(n)} = \rho_j(n) \frac{v_j}{a_n}.$$

□

Identity (20) is the useful form for the mean, since the weights σ_k are dominated by their first few terms. We make no attempt to optimise the analytic threshold; to join it directly to the exact range through 650, throughout this section we fix

$$n_0 := 651, \quad K := 40, \quad M := 1.5604.$$

For $1 \leq k \leq K$ and $n \geq n_0$, condition (17) holds because $9k(n-k) - (k+1)^3$ is concave in k and is positive at both endpoints $k=1$ and $k=40$ (at the latter, $9 \cdot 40 \cdot 611 = 219960 \geq 41^3 = 68921$). Thus Lemma 4.5 gives $\rho_k(n) \geq 1$; and Lemma 4.4 gives $\rho_k(n) \leq P_k := \exp\left(\frac{5k}{2n} + \frac{k^3}{2n^2} + \frac{k^2}{n^2}\right)$, each exponent decreasing in n , so the value at $n = n_0$ is uniform.

Theorem 5.2 (two-sided effective mean). *For every $n \geq 651$,*

$$1.4819 \leq \mathbf{E}[c_n] \leq 1.538952.$$

Proof. Lower bound. Discard from (20) all terms with $k > K$ (they are positive). We need a_{n-k}/a_n from below: Theorem 4.6 gives $a_{n-k} \geq 1.647767$ (valid for every index ≥ 2 , and $n-k \geq 611$ here) and Theorem 4.7 gives $a_n \leq 17/10$. Hence, using $\rho_k \geq 1$ in (21),

$$\mathbf{E}[c_n] \geq 1 + \frac{1.647767}{1.7} \sum_{k=1}^K \left(1 - \frac{k}{n}\right) w_k \geq 1 + \frac{1.647767}{1.7} \left(\sum_{k=1}^K w_k - \frac{1}{n_0} \sum_{k=1}^K k w_k \right) \geq 1.4819;$$

note $\sum_{k \leq K} k w_k \leq 1$, so the bracket is decreasing in $1/n$ and $n = n_0$ is the worst case.

Upper bound. Split (20) at $k = K$. For the head, use $1 - k/n \leq 1$, $\rho_k \leq P_k$, $a_{n-k} \leq 17/10$ (as $n - k \geq 611 \geq 81$) and $a_n \geq 1.647767$:

$$\sum_{k=1}^K \sigma_k(n) \leq \frac{1.7}{1.647767} \sum_{k=1}^K w_k P_k \leq 0.518390291.$$

For the tail, $\sigma_k(n) = q_k(n) a_{n-k}/a_n \leq q_k(n) \cdot \frac{5/2}{1.647767}$ by Corollary 4.8 and Theorem 4.6, and Lemma 4.3 bounds q_k . Splitting at $k = n/2$: for $K < k \leq n/2$ we have $(n-k)^{-3/2} \leq 2^{3/2} n^{-3/2}$ so $q_k \leq 2^{3/2}/(\sqrt{2\pi} k^{5/2})$, and $\sum_{k > K} k^{-5/2} \leq \frac{2}{3} K^{-3/2}$; for $n/2 < k \leq n-1$ we have $k^{-5/2} \leq 2^{5/2} n^{-5/2}$ so $q_k \leq 2^{5/2}/(\sqrt{2\pi} n (n-k)^{3/2})$, and $\sum_{r \geq 1} r^{-3/2} = \zeta(3/2)$. Hence

$$\sum_{k=K+1}^{n-1} \sigma_k(n) \leq \frac{5/2}{1.647767} \cdot \frac{1}{\sqrt{2\pi}} \left(2^{3/2} \cdot \frac{2}{3} \cdot 40^{-3/2} + \frac{2^{5/2} \zeta(3/2)}{n_0} \right) \leq 0.018251379,$$

both bracket terms being nonincreasing in n . Adding, $\mathbf{E}[c_n] \leq 1 + 0.518390291 + 0.018251379 = 1.536641670 \leq 1.538952$. □

Corollary 5.3. $\mathbf{E}[c_j] \leq M = 1.5604$ for every $j \geq 41$.

Proof. For $41 \leq j \leq 700$ this is an exact integer computation; the maximum over that range is $\mathbf{E}[c_{41}] = 1.560354926 \dots$. For $j \geq 651$ it follows from Theorem 5.2, since $1.538952 \leq M$. The ranges overlap. The split at $K = 40$ is forced rather than chosen: $\mathbf{E}[c_{40}] = 1.561838942 \dots > M$. □

We can now bound the second moment. The identity (19) expresses $\mathbf{E}[c_n(c_n - 1)]$ as a τ -weighted sum of $\mathbf{E}[c_j]$ over all smaller j , and the weights τ_j for large j are not individually accessible to any of the estimates above. The following step removes them.

Theorem 5.4 (effective second moment). *For every $n \geq 651$, $\mathbf{E}[c_n^2] \leq 2.9123$.*

Proof. Split (19) at $j = K$ and apply Corollary 5.3 to the tail:

$$\mathbf{E}[c(c-1)] \leq \sum_{j=1}^K \tau_j \mathbf{E}[c_j] + M \sum_{j=K+1}^{n-1} \tau_j = \sum_{j=1}^K \tau_j (\mathbf{E}[c_j] - M) + M \sum_{j=1}^{n-1} \tau_j.$$

This is the tail elimination: by (18) the last sum equals $\mathbf{E}[c_n] - \tau_0 = \mathbf{E}[c_n] - 1/a_n$, so the entire uncontrolled tail has been replaced by quantities already bounded, and only the forty head terms remain explicit. Using $\tau_j = \rho_j v_j / a_n$,

$$\mathbf{E}[c(c-1)] \leq \frac{1}{a_n} \left[\sum_{j=1}^K \rho_j(n) v_j (\mathbf{E}[c_j] - M) - M \right] + M \mathbf{E}[c_n].$$

The summands change sign: $\mathbf{E}[c_j] - M < 0$ exactly for $j \in \{1, 2\}$ and $\mathbf{E}[c_j] - M > 0$ for $3 \leq j \leq K$. Bounding each in its own direction — $\rho_j \leq P_j$ and v_j from above on the positive terms, $\rho_j \geq 1$ and v_j from below on the two negative terms — gives

$$\sum_{j=1}^K \rho_j(n) v_j (\mathbf{E}[c_j] - M) \leq U := -0.1873399.$$

Here U is the outward-rounded value of the explicit forty-term rational expression built from the exact values $A(j)$ and $\mathbf{E}[c_j]$, together with the one-sided exponential enclosures of Appendix A.1. Since $U - M = -1.7477399 < 0$, the bracket is negative, and dividing a negative quantity by a_n is maximised by taking a_n as *large* as possible; Theorem 4.7 supplies $a_n \leq 17/10$, which is therefore the correct one-sided input. With $\mathbf{E}[c_n] \leq B := 1.538952$ from Theorem 5.2,

$$\begin{aligned} \mathbf{E}[c_n^2] &= \mathbf{E}[c(c-1)] + \mathbf{E}[c_n] \\ &\leq \frac{U - M}{17/10} + (M + 1)B = \frac{30942660571}{10625000000} = 2.9122504\dots \leq 2.9123. \end{aligned}$$

□

Remark 5.5. The bivariate exponential generating function for forests is $\exp(uU(z))$, where $T = ze^T$ and $U = T - T^2/2$; u marks components. At $z = e^{-1}$ one has $T = 1$ and $U = 1/2$, so evaluation and differentiation in u give $1 + \sum_{j \geq 1} v_j = \sqrt{e}$ and $\sum_{j \geq 1} v_j \mathbf{E}[c_j] = \frac{1}{2}\sqrt{e}$. The square-root expansion at this point gives the limiting probability generating function $u \exp((u-1)/2)$ [3, Chapter VII]; hence $\mathbf{E}[c_n] \rightarrow 3/2$ and $\mathbf{E}[c_n^2] \rightarrow 11/4$.

Now feed (19) an arbitrary uniform tail bound $\mathbf{E}[c_j] \leq M'$ into the same tail elimination. Reusing M' as a bound for the mean as well would give the coarser $M'^2 + \frac{1}{2}$. Retaining the mean limit $\mathbf{E}[c_n] \rightarrow 3/2$, however, gives the sharper limiting bound $\mathbf{E}[c^2] \leq 2 + M'/2$, again exact at $M' = 3/2$; thus the limiting calculation permits $M' \leq 1.98$. The finite majorants used above are more restrictive: at $n = 651$ they close for $M' = 1.6971$ but not for $M' = 1.6972$. In particular, the crude choice $M' = 2$ gives the bound 3.1626 and cannot close the proof.

6. THE DEGREE-SQUARE BOUND

The last input to (12) is a lower bound on $\mathbf{E}[D_n]$ matching the term $5n$ to within a constant. Nothing in this section uses Sections 3 or 5.

Theorem 6.1. $\mathbf{E}[D_n] \geq 5n - \frac{29}{2}$ for every $n \geq 24$.

The onset at 24 is sharp: exact computation shows that the inequality fails for every $7 \leq n \leq 23$.

The proof occupies the rest of the section. Throughout put

$$(22) \quad \varphi_k := n g(k) - \left(5n - \frac{29}{2}\right) k^{k-2},$$

with g as in (10), and read k^{k-4} as a rational power ($1^{-3} = 1$, $2^{-2} = 1/4$, $3^{-1} = 1/3$).

Lemma 6.2 (φ -decomposition). *For every $k \geq 1$,*

$$\varphi_k = k^{k-4}h(k), \quad h(k) := \frac{29}{2}k^2 - 11nk + 6n,$$

and, for $n \geq 1$,

$$(23) \quad \mathbf{E}[D_n] \geq 5n - \frac{29}{2} \iff \sum_{k=1}^n \binom{n-1}{k-1} \varphi_k A(n-k) \geq 0.$$

Moreover h is an upward parabola with $h(1) = \frac{29}{2} - 5n < 0$ for $n \geq 3$, and its larger root satisfies $r_+ < 22n/29$; hence $\varphi_k \geq 0$ for every $k \geq 22n/29$.

Proof. By (10), for $k \geq 4$, $g(k)/k^{k-4} = k^2 + 3k(k-2) + (k-2)(k-3) = 5k^2 - 11k + 6$, so $\varphi_k = k^{k-4}[n(5k^2 - 11k + 6) - (5n - \frac{29}{2})k^2] = k^{k-4}h(k)$; the closed form of g also holds at $k = 1, 2, 3$, and the identity is checked there directly. Equivalence (23) follows by subtracting $(5n - \frac{29}{2})A(n)$ from (11) and using (2). For the root, $h(22n/29) = \frac{29}{2} \cdot \frac{484n^2}{841} - \frac{242n^2}{29} + 6n = \frac{242n^2}{29} - \frac{242n^2}{29} + 6n = 6n > 0$, and h is increasing beyond its vertex $11n/29 < 22n/29$, so $r_+ < 22n/29$. $\square$

The terms of (23) with k close to n are the large ones, and by Lemma 6.2 they are positive. The plan is therefore: keep the seventeen terms $k = n - j$, $0 \leq j \leq 16$; discard all other nonnegative terms; and majorise the negative ones, which all have $k < 22n/29$.

Lemma 6.3 (kept block). *Let $n \geq n_1 := 500$, $J := 16$. For $0 \leq j \leq J$ write $\varphi_{n-j} = (n-j)^{n-j-2}\beta_j(n)$ with $\beta_j(n) = \frac{29}{2} - \frac{11n}{n-j} + \frac{6n}{(n-j)^2}$. Then*

$$\binom{n-1}{j} \varphi_{n-j} A(j) \geq n^{n-2} \cdot \frac{A(j)}{j!} e^{-j} \left(1 - \frac{j}{n_1}\right)^{j-1} \beta_j^{\text{lo}}, \quad \beta_j^{\text{lo}} := \frac{29}{2} - \frac{11n_1}{n_1 - j} > 0,$$

and consequently

$$\sum_{j=0}^J \binom{n-1}{j} \varphi_{n-j} A(j) \geq n^{n-2} \cdot \text{POS}, \quad \text{POS} := \sum_{j=0}^{16} \frac{A(j)}{j!} e^{-j} \left(1 - \frac{j}{500}\right)^{j-1} \beta_j^{\text{lo}} \geq 5.69291.$$

Moreover these seventeen indices lie outside the negative range: $n - 16 > 22n/29$ for $n \geq 500$.

Proof. The identity $\varphi_{n-j} = (n-j)^{n-j-2}\beta_j(n)$ is (22) with $k = n - j$, using (10). Since $n/(n-j)$ decreases in n and the term $6n/(n-j)^2$ is positive, $\beta_j(n) \geq \beta_j^{\text{lo}}$ for $n \geq n_1$; positivity of β_j^{lo} for $j \leq 16$ is a finite rational check. Next, $\binom{n-1}{j} = \frac{(n-1)\cdots(n-j)}{j!} \geq \frac{(n-j)^j}{j!}$, so

$$\binom{n-1}{j} (n-j)^{n-j-2} \geq \frac{(n-j)^{n-2}}{j!} = \frac{n^{n-2}}{j!} \left(1 - \frac{j}{n}\right)^{n-2},$$

Furthermore,

$$(1 - j/n)^{n-2} = (1 - j/n)^{n-j-1} (1 - j/n)^{j-1} \geq e^{-j} (1 - j/n)^{j-1}.$$

Indeed, $(1 - j/n)^{-(n-j-1)} = (1 + \frac{j}{n-j})^{n-j-1} \leq e^{j(n-j-1)/(n-j)} \leq e^j$ by $1 + x \leq e^x$. Finally $(1 - j/n)^{j-1}$ is nondecreasing in n for $j \geq 1$, so it is at least its value at n_1 ; for $j = 0$ every factor is 1. For the last statement, $n - 16 > 22n/29 \iff 7n/29 > 16 \iff n > 66.3$. $\square$

Lemma 6.4 (discarded block). *Let $n \geq n_1 = 500$ and $K_0 := 30$. Then*

$$\sum_{k: \varphi_k < 0} \binom{n-1}{k-1} |\varphi_k| A(n-k) \leq n^{n-2} \cdot \text{NEG},$$

$$\text{NEG} := \frac{17}{10} \text{HEAD} + M_1 + M_2 + M_3 + \text{FAR} \leq 5.37050,$$

where $\text{EU}(t)$ denotes a rational upper bound for e^{-t} (Appendix A), $\lambda := \frac{11 \cdot \frac{17}{10}}{\sqrt{2\pi}}$, and

$$\begin{aligned} \text{HEAD} &:= \sum_{k=1}^{30} \frac{(11k-6)k^{k-4}}{(k-1)!} \text{EU}\left(\frac{k(498-k)}{500}\right), \\ M_1 &:= \lambda \left(\frac{500}{440}\right)^{5/2} \sum_{30 < k \leq 60} k^{-5/2}, \\ M_2 &:= \lambda \left(\frac{500}{400}\right)^{5/2} \sum_{60 < k \leq 100} k^{-5/2}, \\ M_3 &:= \lambda 2^{5/2} \sum_{k > 100} k^{-5/2}, \\ \text{FAR} &:= \frac{\lambda}{n_1^{3/2}} \left[5^{5/2} \left(\frac{1}{10} + \frac{1}{n_1}\right) + \left(\frac{50}{9}\right)^{5/2} \left(\frac{1}{10} + \frac{1}{n_1}\right) \right. \\ &\quad \left. + \left(\frac{290}{49}\right)^{5/2} \left(\frac{17}{290} + \frac{1}{n_1}\right) \right]. \end{aligned}$$

Proof. Every negative term has $k < 22n/29$ by Lemma 6.2, whence $n - k > 7n/29 \geq 120$, and

$$|\varphi_k| = k^{k-4} (11nk - 6n - \frac{29}{2}k^2) \leq n(11k-6)k^{k-4}.$$

Head, $1 \leq k \leq 30$. Here $n - k \geq 470 \geq 81$, so Theorem 4.7 gives $A(n-k) \leq \frac{17}{10}(n-k)^{n-k-2}$. With $\binom{n-1}{k-1} \leq n^{k-1}/(k-1)!$ and $(n-k)^{n-k-2} = n^{n-k-2}(1-k/n)^{n-k-2} \leq n^{n-k-2}e^{-k(n-k-2)/n}$,

$$\frac{\binom{n-1}{k-1}|\varphi_k|A(n-k)}{n^{n-2}} \leq \frac{17}{10} \cdot \frac{(11k-6)k^{k-4}}{(k-1)!} e^{-k(n-k-2)/n}.$$

Since $k(n-k-2)/n = k - k(k+2)/n$ is nondecreasing in n and e^{-t} is decreasing, the exponential is at most its value at $n = 500$, which is $\text{EU}(k(498-k)/500)$.

Mid and far, $k > 30$. Here $11nk - 6n \leq 11nk$, so $|\varphi_k| \leq 11n k^{k-3}$, and $k \binom{n}{k} = n \binom{n-1}{k-1}$ gives

$$\binom{n-1}{k-1} |\varphi_k| A(n-k) \leq 11 \binom{n}{k} k^{k-2} A(n-k) \leq 11 \cdot \frac{17}{10} \binom{n}{k} k^{k-2} (n-k)^{n-k-2},$$

using Theorem 4.7 again ($n-k > 120 \geq 81$). By Lemma 4.2 the powers cancel exactly:

$$\frac{\binom{n}{k} k^{k-2} (n-k)^{n-k-2}}{n^{n-2}} \leq \frac{1}{\sqrt{2\pi}} \cdot \frac{n^{5/2}}{k^{5/2} (n-k)^{5/2}}.$$

For $30 < k \leq 60$ we have $n/(n-k) \leq n/(n-60) \leq 500/440$; for $60 < k \leq 100$, $\leq 500/400$; for $100 < k \leq n/2$, ≤ 2 ; giving M_1, M_2, M_3 after bounding $\sum k^{-5/2}$ by the corresponding integral. For $n/2 < k < 22n/29$ we split at $k \leq 3n/5$, $k \leq 7n/10$, $k < 22n/29$, where respectively $n-k \geq 2n/5$, $3n/10$, $7n/29$ and $k \geq n/2$, $3n/5$, $7n/10$, so that $\frac{n^{5/2}}{k^{5/2}(n-k)^{5/2}}$ is at most $5^{5/2}n^{-5/2}$, $(\frac{50}{9})^{5/2}n^{-5/2}$, $(\frac{290}{49})^{5/2}n^{-5/2}$; the numbers of integers in the three blocks are at most $n/10 + 1$, $n/10 + 1$, $17n/290 + 1$. Each resulting bound is a decreasing function of n , so its value at $n_1 = 500$ majorises. Summing gives FAR. $\square$

Proof of Theorem 6.1. For $24 \leq n \leq 650$ the inequality is an exact integer computation, in the form $2 \sum_F D(F) \geq (10n-29)A(n)$. For $n \geq 500$, Lemmas 6.3 and 6.4 give

$$\sum_{k=1}^n \binom{n-1}{k-1} \varphi_k A(n-k) \geq n^{n-2} (\text{POS} - \text{NEG}) \geq n^{n-2} (5.69291 - 5.37050) > 0,$$

and (23) concludes. The two ranges overlap on $500 \leq n \leq 650$. $\square$

7. PROOF OF THEOREM 1.2

7.1. The adjacent orbit. For adjacent pairs the two peeling recurrences can simply be compared term by term.

Lemma 7.1 (termwise comparison). $2(n-2)N_{\text{adj}}(n) < 3N_e(n)$ for every $n \geq 3$.

Proof. Compare (4) and (3) term by term at matched $k \geq 3$ (the extra term $k = 2$ of N_e is nonnegative and is discarded). Using (1), the ratio of the k -th term on the left to the k -th term on the right is

$$\frac{2(n-2)\binom{n-3}{k-3} \cdot 3k^{k-4}}{3\binom{n-2}{k-2} \cdot 2k^{k-3}} = \frac{(n-2)\binom{n-3}{k-3}}{k\binom{n-2}{k-2}} = \frac{(n-2)(n-3)!(k-2)!}{(k-3)!(n-2)!k} = \frac{k-2}{k} < 1.$$

Every matched term is positive, and the $k = 2$ term on the right is positive as well, so the summed inequality is strict. $\square$

Proposition 7.2. If $n \geq 12$ and $\mathbf{E}[c_n] \leq 2$, then the inequality (ADJ_n) is strict.

Proof. (ADJ_n) is $N_{\text{adj}} \leq N_e \cdot (N_e/A)$, so by Lemma 7.1 and (9) it suffices that $\frac{3}{2(n-2)} \leq \frac{n-\mu}{\binom{n}{2}}$, i.e. $3n(n-1)/2 \leq 2(n-2)(n-\mu)$. With $\mu \leq 2$ the right side is at least $2(n-2)^2$, and $3n(n-1)/2 \leq 2(n-2)^2$ is equivalent to $n^2 - 13n + 16 \geq 0$, whose larger root is $(13 + \sqrt{105})/2 = 11.6235\dots$; so the condition holds exactly for $n \geq 12$. $\square$

Proof of (ADJ_n) for all $n \geq 3$. For $n \geq 651$, Theorem 5.2 gives $\mathbf{E}[c_n] \leq 1.538952 < 2$, so Proposition 7.2 applies. For $3 \leq n \leq 1200$, strict (ADJ_n) is verified in exact integers. The ranges overlap. $\square$

7.2. The disjoint orbit.

Proof of (DIS_n) for all $n \geq 4$. For $n \geq 651$, combine Proposition 3.1 with Theorem 6.1 ($\mathbf{E}[D] - 5n \geq -\frac{29}{2}$), Theorem 5.2 ($1.4819 \leq \mu \leq 1.538952$, in particular $1 \leq \mu \leq 2$ as required for the bound on R), and Theorem 5.4 ($\mathbf{E}[c^2] \leq 2.9123 \leq 2.99$). Since Δ in (12) is increasing in μ on $\mu > 0$ through the term $9\mu + \mu^2$, while R is bounded using $\mu \leq 2$, the lower bound on μ is the correct input in the first place and the upper bound in the second. Hence for $n \geq 651$

$$\Delta(n) \geq -\frac{29}{2} + 2 + 9(1.4819) + (1.4819)^2 - 2.99 - \frac{22}{650} = \frac{12065893}{1300000000} = 0.00928\dots > 0,$$

and Proposition 3.1 gives (DIS_n) . Using the sharper $\mathbf{E}[c^2] \leq 2.9123$ in place of 2.99 raises the margin to 0.0870\dots For $4 \leq n \leq 1200$, strict (DIS_n) is verified in exact integers. The ranges overlap. $\square$

Proof of Theorem 1.2. For $n = 2$ there is only one edge and the statement is vacuous. For $n \geq 3$, $\text{Aut}(K_n) = S_n$ acts transitively on edges and on each of the two orbits of unordered pairs of distinct edges (adjacent, disjoint), so $\mathbf{P}(e \in F_n) = N_e/A$ and $\mathbf{P}(e, f \in F_n)$ equals N_{adj}/A or N_{dis}/A according to the orbit of $\{e, f\}$. The assertion is therefore exactly (ADJ_n) together with (DIS_n) , both of which are proved above. $\square$

Corollary 7.3. For every $n \geq 3$ and every pair of distinct edges e, f of K_n ,

$$\mathbf{P}(e, f \in F_n) < \mathbf{P}(e \in F_n)\mathbf{P}(f \in F_n).$$

Proof. The adjacent proof is strict by Lemma 7.1; the disjoint proof has $\Delta(n) > 0$ analytically and is checked strictly over the finite range. $\square$

8. CONCLUDING REMARKS

Tang and Zhang’s fixed-component identities, and their later spanning-tree generalization, show that degree moments already encode p-NC when the edge count is fixed [15, 16]. Proposition 3.1 extends that dictionary to the unconditioned forest: mixing the component strata introduces $\text{Var}(m)$, or equivalently the first two moments of c . The reduction remains exact, and the order- n^{-2} cancellation in the original edge counts becomes a constant-scale moment inequality. The substantial work here is the effective control of those fluctuating-component terms.

The effective constants of Sections 4–6 are scaffolding for the present proof and we make no claim that they are sharp; the limits $a_n \rightarrow \sqrt{e}$ [11], $\mathbf{E}[c_n] \rightarrow 3/2$ and $\mathbf{E}[c_n^2] \rightarrow 11/4$ show that each is within a few percent of the truth, and Remark 5.5 derives the component limits and explains why even a few percent of slack matters.

The natural structural question is whether a version of the marking identities of Lemma 5.1 survives for other host graphs through the weighted Matrix–Tree theorem. Here they reduce to Cayley factors, the first point at which completeness is essential. Extending that reduction, rather than further optimising the finite threshold, appears the more promising route toward Conjecture 1.1. Its general finite-graph and multigraph forms, and full negative association of the uniform forest measure even for K_n , remain open.

DATA AND CODE AVAILABILITY

The exact verification program is included with the arXiv submission as an ancillary file and is maintained in the companion repository at <https://github.com/agupta/uniform-forests-complete-graphs>.

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APPENDIX A. THE VERIFICATION PROGRAM

The accompanying program `tests/check.py` is a single standalone file using only the Python standard library and only `int` and `fractions.Fraction` arithmetic. No floating-point number decides any test; decimal strings are produced by integer arithmetic and are display-only. Running `python3 tests/check.py -nmax 700` establishes every finite assertion the proof of Theorem 1.2 depends on; the larger run `-nmax 1200` additionally establishes Computation A.2 over its full range. The target `make check-full` also sets `-ncross 1200`, extending the independent pair-count comparison across that range. Runtime is machine-dependent.

A.1. Rational enclosures. Every estimate in Sections 4–7 terminates in a finite rational expression involving e , π , $\sqrt{2\pi}$ and $\zeta(3/2)$. These are enclosed as follows.

- $E_{\text{lo}} := \sum_{i=0}^{24} 1/i! \leq e \leq E_{\text{lo}} + 2/25! =: E_{\text{hi}}$, the tail estimate being $\sum_{i \geq 25} 1/i! = \frac{1}{25!} \sum_{r \geq 0} \frac{1}{26 \cdot 27 \cdots (25+r)} \leq \frac{1}{25!} \sum_{r \geq 0} 26^{-r} \leq 2/25!$;
- for rational $x \geq 0$ write $x = a + s$ with $a = \lfloor x \rfloor$ and $s \in [0, 1)$. Then, with $N = 32$,

$$E_{\text{lo}}^a \sum_{i=0}^{N-1} \frac{s^i}{i!} \leq e^x \leq E_{\text{hi}}^a \left(\sum_{i=0}^{N-1} \frac{s^i}{i!} + \frac{s^N}{N!} \cdot \frac{N+1}{N} \right),$$

the remainder estimate being $\sum_{i \geq N} \frac{s^i}{i!} \leq \frac{s^N}{N!} \sum_{r \geq 0} \left(\frac{s}{N+1}\right)^r \leq \frac{s^N}{N!} \cdot \frac{N+1}{N}$, valid because $s < 1$. We write $\text{EU}(t)$ for the resulting rational upper bound on e^{-t} , $t \geq 0$ rational;

- square roots are enclosed by integer square roots at 40 decimal digits; π by Machin's formula with an explicit alternating-series remainder; and $\zeta(3/2) = \sum_{j \geq 1} j^{-3/2}$ by an exact partial sum to $J = 20000$ plus the midpoint tail

$$\sum_{j > J} j^{-3/2} \leq \int_{J+1/2}^{\infty} x^{-3/2} dx = \frac{2}{\sqrt{J+1/2}},$$

which is legitimate because $x^{-3/2}$ is convex, so that $f(j) \leq \int_{j-1/2}^{j+1/2} f$ for each j .

Each enclosure is used only in the direction in which it is valid, and each is self-tested by the program.

Remark A.1. Several constants have margins of order 10^{-8} , so the enclosure must be both one-sided and tight. The far block of Theorem 4.7, for example, clears its printed bound by about $5 \cdot 10^{-10}$. The integer-part reduction above has relative width below 10^{-12} on the test grid covering $0 \leq x \leq 60$; the verifier asserts this directly.

A.2. What the program checks. A — definition-level controls. All $2^{\binom{n}{2}}$ edge subsets of K_n are enumerated for $2 \leq n \leq 7$ and the sequences A , $\sum_F c$, $\sum_F c(c-1)$, $\sum_F D$, N_e , N_{adj} , N_{dis} are compared against Propositions 2.3–2.5; $A(0..10)$ is compared against OEIS A001858. Spanning trees are enumerated separately for $3 \leq n \leq 7$, reproducing Observation 1.3 exactly, including $T_{\text{dis}}T = T_e^2$. One check is a negative control, asserting that the comparison used for (ADJ_n) is capable of failing.

B — the exact integer seam. The four sequences A , $\sum_F c$, $\sum_F c(c-1)$, $\sum_F D$ are built from (2) and (11), and the three pair counts are then obtained from (6)–(8); each of the three divisions is asserted to be exact, which is a strong internal consistency test. Independently, the pair counts are recomputed from (3)–(5) through $n = 400$ by default and the two derivations are required to agree; **check-full** extends this comparison through $n = 1200$. The section then verifies strict (ADJ_n) and (DIS_n) , monotonicity of both finite ratio sequences, the exact identity

$$4\Delta(n)A(n)^2 = n(n-1)(n-2)(n-3)(N_e(n)^2 - N_{\text{dis}}(n)A(n)),$$

the component-moment form (12), a negative control omitting $\text{Var}(m)$, and every finite range used above.

C — arithmetic of the analytic majorants. Every rational constant appearing in Sections 4–7 is recomputed from the displayed majorants using the enclosures of Appendix A.1. This checks their arithmetic, not their derivation; the inequalities producing the majorants are proved in the text. This section does not depend on the length of the exact seam.

Each statement of the note that rests on a finite verification can be located in the program's output by tag:

statement	tags
Theorem 4.6	B/a_n>=1.647767, C/ThmL
Theorem 4.7	B/a_80>17/10, B/a_n<=17/10, C/ThmU head middle far total
Corollary 4.8	B/a_n<=5/2
Theorem 5.2	C/ThmMean lower head tail
Corollary 5.3	B/E[c_j]<=1.5604, B/split-at-40-is-forced
Theorem 5.4	C/ThmSecondMoment head U, C/ThmSecondMoment
Theorem 6.1	B/E[D]>=5n-29/2, C/ThmD betaL POS NEG
Proposition 7.2	B/n^2-13n+16-threshold
(ADJ_n) , (DIS_n) , Corollary 7.3	B/ADJ strict, B/DIS strict
	B/exact Delta-gap identity, C/assembly

A.3. Output of the exact seam.

Computation A.2. For $3 \leq n \leq 1200$: (ADJ_n) is strict, with largest ratio $N_{\text{adj}}A/N_e^2 = 7/9$ attained at $n = 3$; the ratio is strictly decreasing throughout this finite range. For $4 \leq n \leq 1200$: (DIS_n) is strict, with largest ratio $N_{\text{dis}}A/N_e^2 = 0.999999305561\dots$ at $n = 1200$, the ratio being strictly increasing in n over the whole range. Also the exact Δ identity displayed above and (12) hold for every $4 \leq n \leq 1200$.

Computation A.3. Selected exact values, truncated after nine decimal places: $a_{651} = 1.655058038\dots$, $\mathbf{E}[c_{651}] = 1.503838989\dots$, $\mathbf{E}[c_{651}^2] = 2.765376527\dots$, $a_{80} = 1.700555369\dots$, $a_4 = 19/8$, $\mathbf{E}[c_{40}] = 1.561838942\dots$, $\mathbf{E}[c_{41}] = 1.560354926\dots$. Each display is asserted by a B/display tag in the verifier.

The theorem needs the exact seam only through $n = 650$; the default proof run to 700 supplies overlap values, and the run to 1200 supplies the finite data quoted above.

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